

① Earliest Deadline First:

#include <stdio.h>

struct Process {

int id, bt, deadline, period, remaining, next_deadline;

};

void edf(struct Process p[], int n){

int i, t = 0;

int max_p = 0;

for(int i = 0; i < n; i++){

if(p[i].period > max_p)

max_p = p[i].period;

}

int hyperperiod = 1;

int found = 0;

while(!found)

found = 1;

for(i = 0; i < n; i++){

if(hyperperiod % p[i].period != 0){

found = 0;

break;

}

if(!found) hyperperiod++;

}

int gantt[1000];

for(i = 0; i < n; i++){

p[i].remaining = 0;

p[i].next_deadline = 0;

}

for(t = 0, t < hyperperiod; t++){

for(i = 0; i < n; i++){

if(t % p[i].period == 0){

p[i].remaining = p[i].bt;

p[i].next_deadline = t + p[i].deadline;

}

}


```

int idx = -1;
int min-deadline = 9999;
for(i = 0; i < n; i++) {
    if(p[i].remaining > 0) {
        if(p[i].next-deadline <
            min-deadline) {
            min-deadline = p[i].next-
                deadline;
            idx = i;
        }
    }
}

```

```

if(idx != -1) {
    gantt[t] = p[idx].id;
    p[idx].remaining--;
}

```

```

} else {
    gantt[t] = 0;
}

```

```

}
printf("Gantt Chart (Hyper-period %d)\n",
    hyperperiod);

```

```

for(i = 0; i < hyperperiod; i++)
    printf("----+");

```

```

printf("\n");

```

```

for(i = 0; i < hyperperiod; i++) {

```

```

    if(gantt[i] == 0) printf("Idle ");

```

```

    else printf("P %d", gantt[i]);

```

```

}

```

```

printf("\n");

```

```

for(i = 0; i < hyperperiod; i++) printf("----+");

```

```

printf("\n");

```

```

for(i = 1; i <= hyperperiod; i++) printf("%d", gantt[i]);

```

```

printf("\n");

```

```

}

```



```

int main() {
    int n, i;
    printf("Enter no. of processes:");
    scanf("%d", &n);
    struct Process p[n];
    for(i=0; i<n; i++) {
        p[i].id = i+1;
        printf("Enter P%d [Burst, Deadline, Period]:", i+1);
        scanf("%d %d %d", &p[i].bt,
            &p[i].deadline, &p[i].period);
    }
    edf(p, n);
    return 0;
}

```

}

O/p:

Enter no. of process: 3

Enter P₁ [Burst, Deadline, Period]: 3 7 20

Enter P₂ [Burst, Deadline, Period]: 2 4 5

Enter P₃ [Burst, Deadline, Period]: 2 8 10

Gantt Chart (Hyper-period: 20)

P ₂	P ₂	P ₁	P ₁	P ₁	P ₃	P ₃	P ₂	P ₂	Idle	P ₂	P ₂	P ₃	P ₃	Idle	P ₂	P ₂	Idle	Idle	Idle	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

② Proportional share scheduling.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
struct process {
```

```
    int id, burst, shares;
```

```
}
```

```
void prop share (process p[], int n, int total_shares,  
                 int total_burst) {
```

```
    srand (time(NULL));
```

```
    int time_elapsed = 0;
```

```
    printf("In Gantt Chart (Total time: %d): \n",  
           total_burst);
```

```
    while (time_elapsed < total_burst) {
```

```
        int lottery = (rand() % total_shares) + 1;
```

```
        int accumulated = 0;
```

```
        for (int i = 0; i < n; i++) {
```

```
            if (p[i].burst > 0) {
```

```
                accumulated += p[i].shares;
```

```
                if (lottery <= accumulated) {
```

```
                    printf("P %d |", p[i].id);
```

```
                    p[i].burst --;
```

```
                    time_elapsed ++;
```

```
                    break;
```

```
                }  
            }  
        }  
    }
```

```
    printf("\n");
```

```
    for (int i = 1; i <= time_elapsed; i++) {
```

```
        printf("    %2d", i);
```

```
    }
```

```
    printf("\n");
```

```
}
```



```

int main() {
    int n, total-shares = 0, total-burst = 0;
    printf("Enter no. of processes:");
    scanf("%d", &n);

    Process p[n];
    for (int i = 0; i < n; i++) {
        p[i].id = i + 1;
        printf("P%d [Burst Shares]: ", i + 1);
        scanf("%d %d", &p[i].burst, &p[i].shares);

        total-shares += p[i].shares;
        total-burst += p[i].burst;
    }

    prop-share(p, n, total-shares, total-burst);
    return 0;
}

```

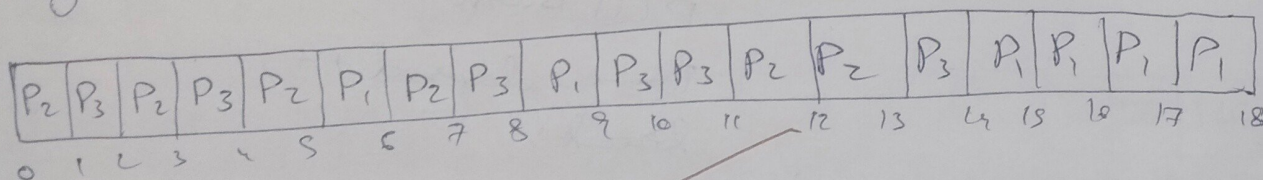
O/P: Enter no. of processes: 3

P₁ [Burst Shares]: 6 1

P₂ [Burst Shares]: 6 2

P₃ [Burst Shares]: 6 3

Gantt Chart (Total time = 18):



Run
27/4/22